MRPL — Standard Operating Procedure

SOP-PM-007: CENTRIFUGAL PUMP MAINTENANCE

Rev. 5 | Effective: 01-Jan-2025 | Area: Process Maintenance

1. PURPOSE

This SOP defines the standard steps for safe and effective preventive and corrective maintenance of centrifugal pumps across all process areas at MRPL. It applies to all maintenance engineers and technicians involved in pump maintenance activities.

2. SCOPE

Applicable to all centrifugal pumps in Process Areas 1–5, including Pump P-101, P-102, P-201, and P-301 series. For high-pressure pumps (>50 bar), refer to SOP-PM-012.

3. SAFETY REQUIREMENTS

3.1 Personal Protective Equipment (PPE)

Mandatory PPE for all pump maintenance activities: Safety helmet, safety glasses (chemical-rated), chemical-resistant gloves (nitrile, min. 0.4mm), fire-retardant coverall, safety boots (steel toe, chemical resistant), face shield when working within 0.5m of flange connections.

3.2 Permit to Work

A valid Permit to Work (PTW) must be obtained before any work commences. PTW must specify: Equipment tag number, work scope, LOTO requirements, gas test requirements, and nearest fire point location. For hot work: additional Hot Work Permit (HWP) required.

3.3 Lock-Out / Tag-Out (LOTO)

LOTO procedure per MRPL-LOTO-001: (1) Notify control room. (2) Stop pump at DCS. (3) Open local isolator switch — lock with personal lock. (4) Isolate suction and discharge valves — apply blinds if required. (5) Isolate seal flush supply. (6) Depressurise casing — open drain valve D-101. (7) Verify zero energy state with multi-meter and pressure gauge. (8) Attach DANGER tag to isolator.

4. PREVENTIVE MAINTENANCE PROCEDURE

4.1 Bearing Inspection and Regreasing

Step 1: Apply LOTO per Section 3.3. Step 2: Remove bearing housing covers. Step 3: Inspect bearing visually for pitting, spalling, discolouration. Step 4: Measure bearing radial play with feeler gauge — acceptable limit 0.02–0.05 mm. Step 5: Remove old grease completely using clean cloth and approved solvent. Step 6: Inspect bearing races for surface damage — replace if any pitting observed. Step 7: Apply fresh grease — Shell Gadus S3 V220C. Fill bearing housing 1/3 to 1/2 capacity. DO NOT over-grease — over-greasing causes bearing temperature rise. Step 8: Replace bearing housing covers — torque to 25 Nm. Step 9: Record in maintenance log: date, grease quantity, condition observations.

4.2 Mechanical Seal Inspection

Step 1: With LOTO applied, drain casing via drain valve D-101. Step 2: Remove coupling guard and flexible coupling element. Step 3: Remove gland plate bolts — note torque values for re-assembly. Step 4: Withdraw mechanical seal assembly carefully — do not scratch shaft. Step 5: Inspect seal faces under good lighting. Acceptable: fine lapping marks only. Reject if: Cracks, chips, heavy scoring, blistering, or erosion visible. Step 6: Inspect O-rings — replace if any deformation, cracking, or chemical attack. Step 7: Measure face wear — reject if face width reduced by >20% from new dimension. Step 8: Clean shaft sleeve and inspect for scoring — replace if deep scratches present. Step 9: Reassemble with new O-rings and spring — do not touch seal faces with bare hands. Step 10: Flush and pressure test seal system at 4 bar for 15 minutes before restart.

4.3 Response to Bearing Temperature Alarm (>85°C)

IMMEDIATE ACTIONS (within 5 minutes of alarm): (1) Notify Shift Supervisor and Control Room Operator. (2) Reduce pump flow by 20% to reduce load. (3) Check bearing temperature trend on DCS historian — is it rising or stable? (4) Dispatch technician for immediate on-site inspection.

ON-SITE INSPECTION: (1) Touch-test bearing housing (with back of hand — do not burn yourself). (2) Listen for bearing noise (grinding, rumbling = bearing failure). (3) Check for visible grease leakage from bearing housing (over-temperature causes grease to liquefy and leak out). (4) Check seal flush rotameter FR-101 — must read 3–5 L/min.

DECISION: If bearing temperature is above 85°C and rising, or any bearing noise is present: Initiate CONTROLLED SHUTDOWN per Section 3.2 and Section 3.3. Switch to standby pump P-101B immediately. Do NOT operate above 95°C — motor trip will activate automatically at 95°C.

5. POST-MAINTENANCE COMMISSIONING

After any maintenance activity, complete the following before returning to service: (1) Remove all LOTO devices — confirm with shift supervisor. (2) Check pump rotation direction (bump test): must be clockwise viewed from drive end. (3) Open suction valve fully, prime pump casing. (4) Start on minimum flow — confirm no unusual noise or vibration. (5) Monitor bearing temperatures for first 30 minutes — must stabilise below 75°C. (6) Confirm seal flush flow at FR-101: 3–5 L/min. (7) Complete maintenance close-out report and sign off PTW.

6. DOCUMENTATION

All maintenance activities must be recorded in the SAP PM system within 24 hours of completion. Required fields: Equipment tag, work order number, work performed (description), parts replaced (part numbers and quantities), next planned maintenance date, engineer signature and employee ID.